

Vision Document for “RentaCar”

Team members:

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1. Introduction

Several years ago, local vehicle rental businesses managed operations using relatively simple Excel spreadsheets, and customers were assigned to cars via a completely manual process. As the business grows, tracking a diverse fleet of vehicles (Sedans, SUVs, Trucks, Luxury cars, etc.) across multiple categories has become increasingly difficult. Car rental operations require managing multiple factors:

- **Vehicle Categories:** Sedans, SUVs, Trucks, and Luxury/Sports cars.
- **Maintenance & Profiles:** Tracking mileage, service availability, and rental history.
- **Customer Demographics:** Regular local customers, long-term corporate clients, and international tourists with different rental duration needs.

Staff need to be able to enter car profiles, manage availability, and view active rentals.

Customers should be able to view available vehicles and easily reserve a car.

RentaCar is a new software tool that will build a centralized booking schedule with vehicles assigned to customers efficiently, offering a simple interface for customer reservations.

2. Positioning

2.1 Problem Statement

The problem of	Managing vehicle rental availability, fleet maintenance status, and allowing clients to book cars.
Affects	System administrators, rental staff, and customers.
The impact of which is	Fleet tracking is highly complex, prone to double-booking, must be manually updated, and changes frequently based on vehicle returns or maintenance delays.
A successful solution would be	A single tool that compiles vehicle availability, integrates business rules for rental durations/pricing, provides a secure database, and offers an intuitive user interface for staff and customers.

2.2 Product Position Statement

For	Car rental companies and their customers.
Who	Need an efficient, error-free way to browse, reserve, and manage vehicle fleets.
The RentaCar system	Is a centralized Car Rental Management Web/Desktop Application.

That	Seamlessly automates rental processing, tracks vehicle status in real-time, and provides quick booking tools.
Unlike	Manual spreadsheets or generic, over-complicated enterprise software.
Our product	Provides a lightweight, high-performance, and straightforward user experience tailored specifically for fast local rental operations.

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Admins	Manage system settings, add/edit/delete employee credentials, and handle configuration.	Setting up the initial software system, managing data backups, and maintaining system security.
Rental Staff	Track and manage car profiles, process pick-ups, and handle returns.	Updating car status (Available, Rented, Maintenance) and reviewing rental agreements.
Customers	Browse live vehicle availability and place rental bookings.	Creating a user account, reserving vehicles, and picking/dropping off vehicles on time.
Developers	Build the application based on the requirements.	Coding system features, fixing software bugs, and ensuring smooth server deployment.
Testers	Validate system behavior using unit and integration tests.	Ensuring booking logic functions perfectly, checking security boundaries, and logging system defects.

3.2 User Environment

- **Users Involved:** Staff handles dozens of bookings per day. Customers interact briefly during reservation or return phases.
- **Platforms:** Web-based interface accessible via desktop (for administration and counter staff) and responsive mobile browsers (for customers).

4. Product Overview

4.1 Product Perspective

RentaCar is an independent, self-contained application incorporating its own relational database. It manages user authentication, fleet tracking, and rental calculations in one integrated package.

4.2 Assumptions and Dependencies

- The application assumes continuous network connectivity to a centralized database server.
- Real-time payment processing is out of scope for this simple version; pricing calculations will happen in-system, but payments are processed manually at the rental counter.

4.3 Needs and Features

No	Problem	Need	Priority	Features	Planned Release
Admin & Staff Features					
1	Fleet expansion or changes.	System must scale easily as new cars are bought or retired.	High	Staff can add, update, or remove vehicles from the DB.	Release 1.0
2	Tracking various vehicle body types.	Sorting cars by operational utility.	Medium	Admin can define specific vehicle categories (SUV, Sedan, Truck).	Release 1.0
3	Vehicles need regular servicing.	Preventing broken cars from being rented.	High	Staff can change vehicle status to "Maintenance" to remove it from customer view.	Release 1.0
4	Dynamic pricing based on classes.	Set base daily rates per vehicle class.	Medium	Admins can configure base pricing tiers.	Release 1.5
Customer Features					
5	Secure login requirements.	Customers and staff need restricted feature access.	High	Role-based secure login screen with credentials provided during registration.	Release 1.0

No	Problem	Need	Priority	Features	Planned Release
6	Finding an available vehicle.	Checking vehicle availability for a specific timeframe.	High	Search and filter tools based on rental dates and vehicle categories.	Release 1.0
7	Reserving a car.	Securing a vehicle before coming to the lot.	High	Customers can pick an available vehicle and submit a reservation block.	Release 1.0
System Rules					
8	Overbooking prevention.	Two customers shouldn't book the same car at the same time.	High	Automated overlap validation rules check vehicle availability prior to confirming reservations.	Release 1.0
9	Rental capacity warning.	Alerting when a vehicle tier is heavily booked.	Low	System warns staff if remaining available inventory drops below 10%.	Release 1.5

4.4 Alternatives and Competition

- **The Status Quo:** Maintaining manual Excel spreadsheets. (Weakness: High risk of double-booking, slow customer service, data corruption risk).
- **Enterprise ERP Software:** Over-engineered commercial software. (Weakness: Extremely high cost, steep learning curve, too heavy for a simple local business).

5. Other Product Requirements

- **Usability:** The UI must be highly intuitive, allowing staff to check out a vehicle in under 3 simple steps.
- **Robustness:** If database operations fail mid-booking, transactions must fully roll back to avoid phantom reservations.
- **Testing Standards:** JUnit or equivalent testing libraries will be leveraged to validate backend database queries and booking validation logic.